

Fawz Haaron

+91-8925126398 | fawzhaaron216@gmail.com | @Fawz-Haaron | @fawz-haaron | @Fawz-Haaron

Education

B.E. Computer Science & Engineering (AI & ML) <i>Relevant Coursework: DSA, OS, DBMS, Computer Networks, ML, Deep Learning</i> CGPA: 7.33 / 10.0	Chennai, India <i>Sep 2024 – Mar 2028</i>
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Technical Skills

Languages: Rust, Python, C, C++, Java, SQL, Bash/Shell, Lua
AI / ML: PyTorch, TensorFlow, ONNX, MediaPipe, OpenCV, BiLSTM, XGBoost, SHAP, MLflow, DVC
Backend & APIs: FastAPI, Axum (Rust), REST, WebSockets, JWT Auth, Celery, Redis, Rate Limiting, Bg Workers
Databases: PostgreSQL, SQLite, Redis; indexing, transactions, query optimization, connection pooling
Systems & Linux: systemd, cgroups, namespaces, process signals, SSH, networking, embedded Linux
Embedded & Hardware: MAVLink, GStreamer, USB/serial protocols, microcontrollers, firmware flashing, SBCs
DevOps & Tooling: Docker, Docker Compose, GitHub Actions (CI/CD), Nginx, Git, Neovim, Cargo, Pyenv, Nix

Experience

Centre for Defence and Space Research (CDSR) <i>Embedded Systems / Drone Software Intern</i> Languages: Rust, Python	Jul 2025 – Jan 2026 <i>Chennai, India</i>
- Built pifeed: modular real-time drone video streaming + telemetry system using GStreamer pipelines, MAVLink packet ingestion, and embedded Linux on SBC hardware. - Parsed MAVLink telemetry (battery, GPS, link quality, sensor state) in Python and exposed structured data via REST for live dashboards. - Implemented configurable H.264 GStreamer encoding pipeline supporting multi-source capture under constrained CPU/memory on embedded targets. - Interfaced with USB/serial hardware, handled firmware flashing workflows, and configured Linux device drivers for drone peripherals.	

Projects

Tethys-Log <i>Rust, GTK4, SQLite, CLI</i> github ↗	2026
- Structured log ingestion, search & visualization tool in Rust with GTK4 UI; inline media viewing, web embeds, code blocks, Vim-motion navigation, and daemon-mode collection. - SQLite-backed storage with regex search, field filtering, configurable retention, and composite index optimization for fast queries at scale.	
TREE-PORT <i>Rust, CLI, content-addressable storage</i> github ↗	2025
CLI time machine for arbitrary directory trees: snapshot-based version history, incremental diffing, content-addressed storage, integrity checksums, and restore workflows — git semantics for non-git directories.	
Operator Fatigue Detection System <i>Python, PyTorch, MediaPipe, BiLSTM, ONNX, FastAPI</i> github ↗	2024
Real-time CV pipeline for operator fatigue detection using MediaPipe + BiLSTM + ONNX for CPU edge deployment. Adaptive Q-learning alerts that personalize fatigue thresholds from operator feedback.	

Notable

- Daily-drives **Arch Linux + Hyprland (Wayland)**; deep experience with systemd, bootloaders, disk encryption, Nix, networking, and kernel tooling. Neovim power user with full Lazy.nvim config (LSP, Mason, Treesitter, custom keymaps).
- Certifications:** Cisco CCNA; NPTEL (Data Science, Networking, AI); CSC (Core Java, ML, Algorithms, System Design, Principle-first Backend)
- LeetCode: 600+ problems solved — **Knight Badge**
- Hackathon Finalist (#3)**, E-Summit 2026 Ideathon at IIITDM Kancheepuram (EdTech track)
- OSS Contributor** and tester at : Black Arch linux, Croc.